INCH-POUND

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MILITARY SPECIFICATION

MIXERS, LIQUID, REVOLVING SHAFT AND AGITATOR TYPES

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric- and pneumatic-motor-driven revolving-shaft and electric-motor-driven agitator-type liquid mixers.

1.2 Classification. The mixers shall be of the following types, classes, and sizes, as specified (see 6.2.1).

Type I - Revolving-shaft mixer
Class 1 - Electric-motor-driven

Size - 1/4 horsepower (hp)
Size - 1/2 hp

Class 2 - Pneumatic-motor-driven

Type II - Agitator-type mixer
Size - 1/4 hp
Size - 1/3 hp

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-22085	- Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment, Requirement 4.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packaging, and Marking.
MIL-STD-1474	- Noise limits for Army Materiel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF LABOR

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29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1901 N. Fort Myer Drive, Arlington, VA 22209.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 153 - Zinc Coating (Hot Dip) On Iron and Steel Hardware.
ASTM A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
ASTM A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality.
ASTM B 26 - Aluminum Alloy Sand Castings.
ASTM B 633 - Zinc On Iron and Steel, Electrodeposited Coatings of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Dimension of Plugs and Receptables - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.4) in accordance with 4.3. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The mixer shall be new and of the contractor's current design. The mixers shall consist of a motor, motor bracket, a shaft assembly, a clamp and bracket mounting device, metal base, transmission case and cover, bearing stand, belt guard, and adjustable can holder.

3.2.1 Measurement system. The US customary system of units (US) or the International System of Units (SI) may be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in the US customary system of units (US). These measurements may be converted to the SI through the use of the conversion factors and methods specified in FED-STD-376.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Fabrication. The end item of equipment shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the end item of equipment, to its maximum rating and capacity, without permanent distortion.

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3.4.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, or other defects. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.4.2 Welding, brazing, and soldering. All welding, brazing, and soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles cracks, gouges, holes, insufficient materials used. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance or repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.4.4 Threads. All machined threads shall conform to FED-STD-H28 and the applicable detailed standards referenced therein.

3.4.5 Lubrications. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.4.6 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The mixer shall comply with the general safety and health requirements promulgated under 29CFR 1910 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.5).

3.4.7 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.4.8 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.5 Gears. All gear and pinion designs selected for use in the end item of equipment shall be manufactured in accordance with AGMA 390.03, AGMA 360.02 and FED-STD-376. Gears and mountings shall be designed and machined to provide a fit-up to minimize backlash.

3.6 Motor. Unless otherwise specified (see 6.2.1), the motor shall conform to NEMA MG-1, as applicable to the type mixer specified. The motor shall be equipped with overload protection and rated for continuous duty operations, and the mixer shall operate from a 115 volt, 60 Hertz, single phase, alternating current source. When specified (see 6.2.1), the electric motor shall be explosion-proof in accordance with NFPA 70.

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3.6.1 Electric power supply cable. The electric power supply cable shall be not less than ten (10) feet in length and shall terminate with a three (3) prong grounding plug in accordance with ANSI C73.11. The three (3) prong plug shall be furnished with a three to two prong outlet adapter, with a ground lead. The electric power supply cable shall include sheathing/covering in accordance with NFPA 70 and 29 CFR 1910. The electric power supply cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the mixer in accordance with NFPA 70.

3.6.2 Switch. The motor shall be equipped with a toggle "off-on" switch mounted on the motor or some adjacent part of the mixer. The switch shall have a current rating of not less than 1-1/2 times the full load current rating of the motor. The switch shall be clearly and permanently marked to indicate the "on" and "off" positions. When specified (see 6.2.1), the switch shall be explosion-proof in accordance with NFPA 70.

3.6.2.1 Fungus resistance. Electrical and electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not weather resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Nameplate. The nameplate and information plate(s) can be incorporated into one or more corrosion-resistant metallic plate(s) at the machine manufacturer's option in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets. The nameplate shall be secured with permanent fasteners at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number
Power input (volts, ampere, phase, Hertz, alternating current (AC)
Horsepower

3.8 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting.

3.9 Type I - revolving-shaft mixer. The type I mixer shall consist of a motor, a propeller shaft, a shaft connector, and a propeller. Each mixer shall start and operate under full-load capacity for not less than 4 hours in an ambient temperature of 70 degrees Fahrenheit (F), ± 5 degrees F, without loss of power, overheating, burning out of any part, permanent distortion, failure of any part, or loosening of fastening devices.

3.9.1 Propeller shaft. The propeller shaft shall be fabricated of type 304 or 316 stainless steel. The shaft shall be of sufficient length to place the propeller within 3 inches of the bottom and sides of a 55-gallon drum. The shaft shall be

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provided with flat side surfaces to seat the setscrews of the shaft connector and the propellers. The shaft shall transmit power to the propellers without torsional vibration or whipping.

3.9.2 Shaft connector. A shaft connector shall be provided to connect the motor shaft to the propeller shaft. When sleeve-type connectors are used, the connector shall be secured to both shafts by setscrews or setscrew and key combinations.

3.9.3 Propeller. The propeller shall be fabricated of type 304 or type 316, stainless steel or aluminum alloy conforming to ASTM B 26, Alloy 355.0, Condition T6, or Alloy 356.0, Condition T6. The propeller shall be a 3-bladed type and shall be secured in a fixed position at the bottom of the shaft by countersunk setscrews. The blade angle shall permit maximum agitation of material without cavitation and shall be consistent with the propeller diameter and motor horsepower.

3.9.4 Mounting device. A clamp-and-bracket mounting device shall be provided with each mixer. The mounting device shall clamp the mixer firmly on the side of a 55-gallon open-head drum, tank, or vessel with shells not more than 1.25 inches thick. The clamp shall lock the mixer securely in an angular off-center mixing position. The bracket shall be adjustable to provide various angular positions of the shaft.

3.9.5 Mixing tank and mounting stand. When specified (see 6.2.1), a mixing tank and mounting stand shall be furnished. The mixing tank shall have a capacity of not less than 50 nor more than 60 gallons and shall be fabricated from corrosion resisting steel sheet conforming to ASTM A 167, type 304 or 316, or carbon steel sheets conforming to ASTM A 366. When carbon steel sheets are used, both the tank and cover shall, after fabrication, be galvanized in accordance with ASTM A 153 or ASTM B 633. The bottom of the tank shall be equipped with a 2-inch outlet fitted with a 2-inch gate valve. The tank shall be provided with either a split cover or a removable cover. The split cover shall have one section hinged to provide an opening to accommodate the shaft of the mixer. The stationary part of the cover shall be bolted or clamped to the top reinforcing ring of the tank and shall be removable at any time. The cover shall have a circular opening of proper size to accommodate the shaft of the mixer. The stand shall support a full tank. A bracket for receiving the clamping device on the mixer shall be mounted on the stand.

3.10 Type I, class 1 - electric-motor-driven.

3.10.1 Size 1/4 hp. The size 1/4-hp mixer shall be equipped with two 4-inch-diameter propellers angled to discharge to the bottom of the container. The upper propeller shall be adjustable on the shaft to within 6 inches of the shaft connector and to within 6 inches of the bottom propeller and shall be secured to the shaft with countersunk setscrews. The mixer shall operate at not less than 1725 revolutions per minute (rpm).

3.10.2 Size 1/2 hp. The size 1/2-hp mixer shall be of a reduction-gear type and shall rotate the propeller shaft at 410 rpm, $\pm$ 10 rpm. The mixer shall be equipped with two 8-inch-diameter propellers angled to discharge to the bottom of the container. The upper propeller shall be adjustable on the propeller shaft to within 6 inches of the shaft connector and to within 6 inches of the bottom propeller and shall be secured to the shaft with countersunk setscrews.

3.11 Type I, class 2 - pneumatic-motor-driven. The type I, class 2, 1/2 hp pneumatic motor driven mixer shall be of a reduction-gear type and shall rotate the propeller shaft at 20 to 400 rpm with a corresponding motor speed 4 to 5 times

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greater. The air motor will develop a minimum of 1/2 hp at 1100 rpm and 60 pounds per square inch (psig). An air muffler shall be provided. The exhaust outlet shall be positioned so that oil or water from the exhaust cannot drop into the material being mixed.

3.11.1 Propeller. The mixer shall be equipped with two 7-inch to 9-inch diameter 3-bladed, square pitch propellers to discharge to the bottom of the container. The upper propeller shall be adjustable on the propeller shaft to within 6 inches of the shaft connector and to within 6 inches of the bottom propeller and shall be secured to the shaft with countersunk setscrews.

3.11.2 Air connection. The air inlet shall have an internal 1/4-inch NPTF thread for attaching the hose.

3.11.3 Quick-disconnect coupling. Unless otherwise specified (see 6.2.1), a 1/4-inch universal quick-disconnect air-line coupling with male and female unit shall be furnished.

3.11.4 Noise limits. When the noise level of the mixer is 85dB(A) or greater, the mixer shall be posted in accordance with MIL-STD-1474. The sign will be modified to indicate the distance from the mixer at which the 85dB(A) limit is met. An appropriate discussion of the noise hazards will be included in the manuals in accordance with MIL-STD-1474.

3.12 Type II - agitator-type mixer. The type II agitator-type mixer shall vibrate a can through a belt and eccentric drive at a speed that will completely mix and blend the contents of the can. All belts shall be provided with a belt guard. When tested as specified herein, the mixer shall operate at its maximum load without loss of power, over-heating, burning out, permanent distortion, or failure of any part.

3.12.1 Can holder. The can holder shall clamp both the top and bottom of the container. The top clamp shall contact the periphery of the container in a manner to reseal a previously opened container. A handcrank or handwheel shall be provided to secure the can in the can holder. Can holders for the size 1/3-hp mixer shall be such that the can can be locked in the right side up, upside down, or sideways position.

3.12.2 Timer. Unless otherwise specified (see 6.2.1), a timer shall be provided for controlling the mixing time. The timer shall be either an automatic mechanical or electric timing device, as specified (see 6.2.1). When specified (see 6.2.1), the timer shall be explosion-proof in accordance with NFPA 70. The timer shall start the mixer without failure and shall stop the mixer at the preset time, $\pm$ 10 percent of full scale.

3.12.3 Size 1/4 hp. The size 1/4-hp mixer shall accommodate containers of from 1/4-pint to 1-gallon capacity. The oscillating or load shaft shall be mounted on bronze sleeve-type bearings. The crank-shaft and spider, or other reciprocating mechanism, shall be mounted on antifriction or sleeve-type bearings and shall be oil-bath lubricated. Moving parts in the transmission shall operate in a bath of oil. When sealed-for-life antifriction bearings are used, oil-bath lubrication shall not be required. The timer shall have an adjustable range from 30 seconds to 15 minutes.

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3.12.4 Size 1/3 hp. The size 1/3-hp mixer shall have a minimum of 1/3 hp, shall accommodate containers of from 1- to 5-gallon capacity, and shall be provided with an adapter for holding a carton of four 1-gallon cans. The eccentric transmission and can holder shaft shall be mounted on friction-type or antifriction-type bearings. When friction-type bearings are used, provisions shall be made for constant lubrication. The timer shall have an adjustable range of from 1 to 30 minutes.

3.13 Workmanship. The mixer and all component parts shall be fabricated free from any defects which could affect reliability or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified in the contract, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all the tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

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4.5 Product examination. Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.2 through 3.5, 3.7 through 3.8, 3.9.1 through 3.11.3, 3.12.1 through 3.12.2, and 3.13. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to identified specified standards, adequacy of markings, proper cleaning, and freedom from defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.7), shall be cause for rejection.

4.6 Tests. Prior to tests, lubricate the mixer with oils and greases. Oils shall be those designated for use in the ambient temperatures at which the test is being performed.

4.6.1 Type I. Mount the mixer on a mixing tank (when provided) or on a 55-gallon open-head drum. Fill the tank or drum with fluid with a viscosity of not less than 120 Krebs units. Operate the mixer continuously for not less than 4 hours under the conditions specified in 3.9. Nonconformance to the requirements of 3.9 shall constitute failure of the test.

4.6.1.1 Horsepower. Attach the pneumatic motor to a calibrated load absorption device. Operate the motor with the adjustable air control valve in its wide-open position and the air pressure regulated to 60 psig. Apply sufficient load to the motor to reduce the speed to 1100 rpm. Failure of the motor to develop 1/2 hp shall constitute failure of this test (see 3.11).

4.6.1.2 Sound level measurement. Operate the pneumatic motor with the adjustable air control valve in its wide-open position and the air pressure regulated to 60 psig and measure the sound level at the operator's position in accordance with MIL-STD-1474 for steady state noise. When the noise level is 85dB(A) or greater the requirements of 3.11.4 shall be met.

4.6.2 Type II.

4.6.2.1 Size 1/4 hp. Clamp a 1-gallon can of paint that has been opened and resealed in the can holder. Operate the mixer for 15 minutes. Failure to completely break up the pigment, settlement and blending with the paint vehicle, or evidence of paint leakage at the lid shall constitute failure of this test. Continue to operate the mixer for 1 hour. Nonconformance to the requirements of 3.12, 3.12.1, and 3.12.3 shall constitute failure of the test.

4.6.2.2 Size 1/3 hp. Clamp a 5-gallon can of paint that has been opened and resealed in the can holder. Operate the mixer with the can holder in the right side up position for 30 minutes. Repeat this test with the can holder locked in the upside down and in the sideways positions. Nonconformance to the requirements of 3.12, 3.12.1, and 3.12.4 shall constitute failure of the test.

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4.6.2.3 Timer. Set the timer for the minimum and then the maximum time specified. Failure of the timer to start the mixer, or failure of the timer to stop the mixer within the specified time, ± 10 percent, of full scale shall constitute failure of this test (see 3.12.2).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Level C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Disassembly. Mixers shall be disassembled in accordance with the recommendation of the manufacturer.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unprotected surfaces. Unpainted exterior metal surfaces of components and parts of the mixers such as bolts, nuts, setscrews, pins, clamps, brackets, or any other unpainted exterior metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.1.1.4 Electric motors and timers. Openings into electric motors and timers shall be sealed with tape conforming to MIL-T-22085, type II. The power cable shall be coiled or hanked. Coiled cable shall be secured with tape specified herein.

5.1.1.5 Pneumatic motors. Interior surfaces of pneumatic motors shall be coated with type P-10 preservative, grade 30, applied through the lubricating system or by feeding the preservative into the air inlet while the motors are operated until preservative appears at exhaust ports. Air inlets and outlets shall be sealed with plastic or metal caps or plugs, as applicable, or covered with barrier material conforming to MIL-B-121, type II, grade A, class 1 or 2 and secured in place with tape conforming to MIL-T-22085, type II.

5.1.1.6 Gears.

5.1.1.6.1 Enclosed gears. Gear housings shall be filled to the operating level with the lubricant recommended for operation and the gears actuated to assure complete coverage.

5.1.1.6.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.1.1.7 Drive belts and pulleys. Belts shall be removed or released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.8 Mixing tank. The cover shall be secured to the mixing tank in a manner to prevent movement or damage and the gate valve placed in the open position.

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5.1.1.9 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.10 Consolidation.

5.1.1.10.1 Type I mixers. The technical publications and the components of each type I mixer, except for the mixing tank and mounting stand, shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c, style optional. The contents shall be immobilized within the box in a manner to prevent movement or damage. The box shall be closed and sealed as specified for method V of the appendix to the box specification.

5.1.1.10.2 Type II mixers. The technical publications and the components removed from the type II mixer shall be consolidated as specified in 5.1.1.10.1.

5.1.2 Level C. Level C preservation of each mixer shall be in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be Level A, B, or Level C, as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Type I mixers furnished without a mixing tank and stand. Type I mixers of like description furnished without a mixing tank or stand shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box specification.

5.2.1.2 Type I mixers furnished with a mixing tank and stand. Type I mixers furnished with a mixing tank and stand shall be individually packed as specified in 5.2.1.1. Blocking, bracing, and anchoring of the contents within the box to prevent movement or damage shall be provided.

5.2.1.3 Type II mixers. Type II mixers shall be packed as specified in 5.2.1.2.

5.2.2 Level B. Mixers preserved as specified in 5.1 shall be packed as specified in 5.2.1, except that boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-640, class 2, style optional, or PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed together does not exceed the gross weight or size limitations of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specifications.

5.2.3 Level C. Level C packing of the mixers shall be in accordance with MIL-STD-1190.

5.3 Marking. All marking shall be inaccordance with MIL-STD-129. Bar Code marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I mixers are intended to be clamped to open-head drums, tanks, or vessels for the mixing of paints, oils, or emulsions. Type II mixers are intended for mixing paints, oils, or emulsions furnished in closed containers.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).
d. First article when required (3.1).
e. Exceptions and additional safety/health requirements when required (see 3.4.6 and 6.5).
f. Motor, if other than that specified (see 3.6).
g. When explosion-proof motors are required (see 3.6).
h. When explosion-proof switches are required (see 3.6.2).
i. Painting, if different (see 3.8).
j. When a mixing tank and mounting stand shall be provided (see 3.9.5).
k. When a quick-disconnect coupling is required (see 3.11.3).
l. When a timer is not required (see 3.12.2).
m. Whether the timer shall be an automatic mechanical or electric timing device (see 3.12.2).
n. When an explosion-proof timer is required (see 3.12.2).
o. First article inspection, if different (see 4.3).
p. Quality conformance inspection, if different (see 4.4).
q. Level of preservation and packing, if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Sampling. Sampling for quality conformance inspection should be in accordance with MIL-STD-105, using Inspection Levels and Acceptable Quality Levels (AQL), as listed, unless 100 percent inspection is specified. The lot size shall be limited to 150 units.

Inspection or Test	Inspection Level	AQL
Product examination (see 4.5)	II	1.0
Operational test (see 4.6)	100 percent	----
Packaging inspection (see 4.7)	S-4	4.0

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6.7 Subject term (key word) listing.

Propeller
Motor
Pneumatic
Can holder
Timer

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Review activity:
DLA-CS

User activities:
Army - ER
Navy - MC
Air Force - 82

Preparing activity:
Army - AL

Project No. 4940-0590

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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